

New York City Airbnb

Data Analysis

A structured analytical framework for understanding listing behavior

48,895

Listings

16

Features

2019

Dataset

Research Question & Objectives

Research Question: How does the number of guest reviews relate to listing price, availability, and other characteristics of a listing?

① Identify Factors

Key variables influencing availability and prices related to number_of_reviews

② Predict Behavior

Whether prices and availability increase or decrease based on review volume

③ Detect Patterns

Anomalies and distributional patterns across boroughs and room types

Dataset Overview

Key Characteristics

Source NYC Airbnb Open Data (Kaggle)

File AB_NYC_2019.csv

Rows 48,895 listings

Columns 16 features

Types Numerical (int64, float64) +
Categorical (str)

Missing Values

name: 16 missing

host_name: 21 missing

last_review / reviews_per_month: 10,052 missing
(listings with no reviews — not errors)

Key Variables

Target: number_of_reviews

Numerical: price, availability_365, reviews_per_month

Categorical: room_type, neighbourhood_group

Data Preparation

1

Data Cleaning

Missing values imputed using median. Duplicate entries identified and removed. Categorical fields standardized.

2

Outlier Detection (IQR)

price: 2,972 outliers | number_of_reviews: 6,021 outliers (12%) | availability_365: 0 outliers

3

Encoding

room_type and neighbourhood_group converted to numeric codes for model compatibility.

4

Log Transformation

Applied to reduce skewness: price 19.12 $\rightarrow$ 0.55, reviews 3.69 $\rightarrow$ 0.36, availability 0.76 $\rightarrow$ -0.20

5

RobustScaler

Applied after log transform. Median $\approx$ 0 for all variables. Resistant to remaining outliers.

Outlier Analysis

Outliers by Variable (IQR Method)



price

2,972

Extreme luxury listings

number_of_reviews

6,021

12% of dataset — significant

availability_365

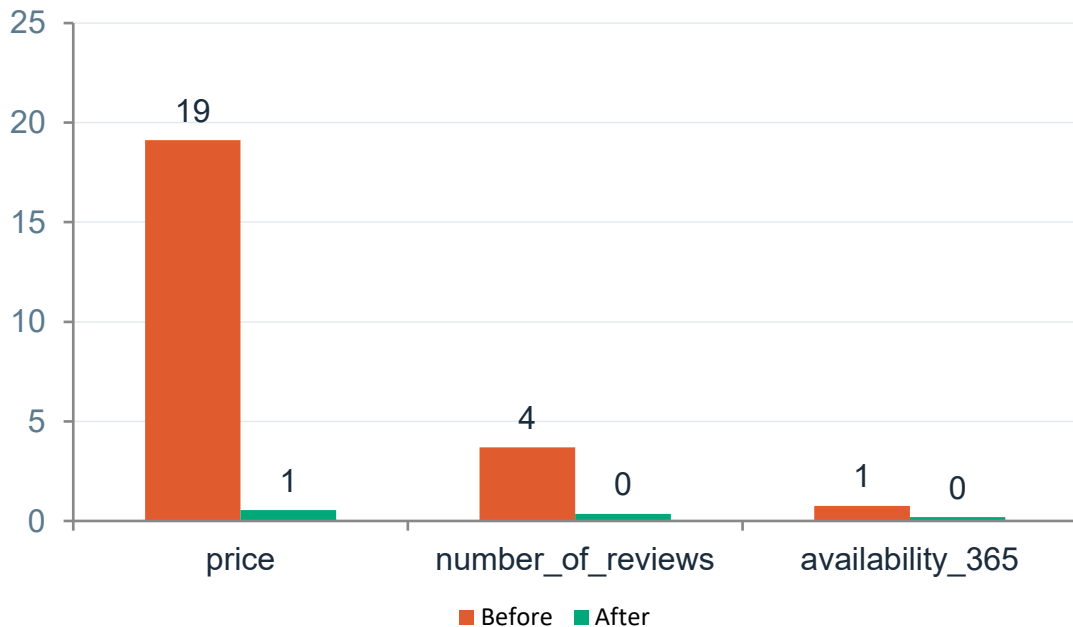
0

No outliers — clean variable

Decision: Use RobustScaler, resistant to outliers in price and reviews

Log Transformation — Skewness Reduction

Skewness Before vs After Log Transform



Variable	Before	After	Result
price	19.12	0.55	✓ Excellent
number_of_reviews	3.69	0.36	✓ Great
availability_365	0.76	-0.20	✓ Near Perfect

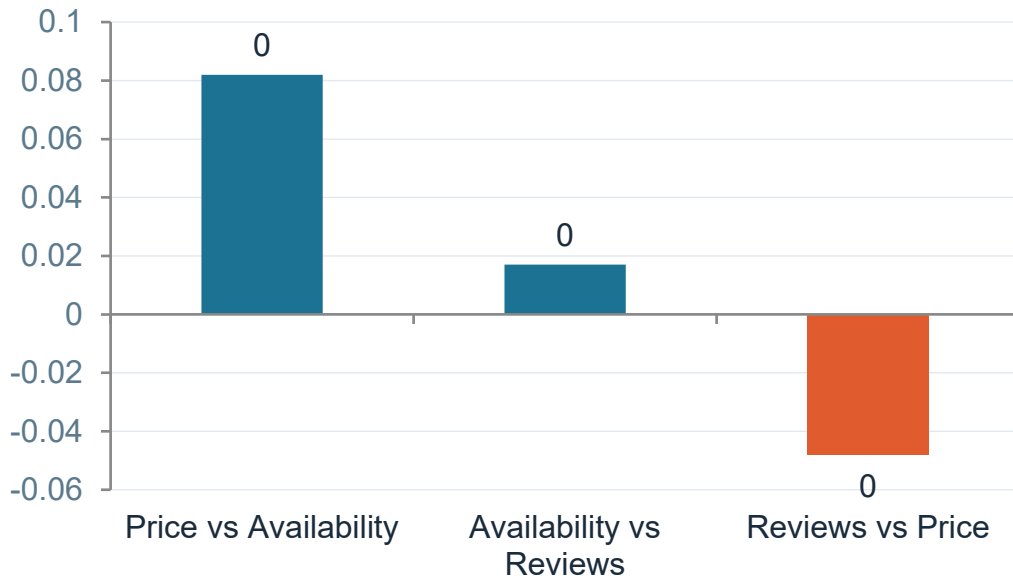
Ideal range: -0.5 to $+0.5$

All 3 variables now within range after transformation.

Log transform applied first → RobustScaler applied second

Correlation Analysis

Pairwise Correlation Coefficients



Overall Correlation

-0.048

Very weak negative

P-Value

2.69e-26

Statistically significant

Price vs Availability

0.082

No meaningful correlation

Availability vs Reviews

0.017

Key Insight: High p-value significance does not equal strong correlation, the relationship is real but negligible.

Exploratory Data Analysis — Key Visualizations

Histograms



price, number_of_reviews, and availability_365, all right-skewed, confirming outlier presence and need for log transformation.

Box Plots



Dots far beyond whiskers confirmed significant outliers in price and reviews. Availability showed clean distribution.

Pair Plot



Scatter matrix of all 3 variables, no clear linear patterns, confirming weak correlations across all pairs.

Correlation Heatmap



Matrix of Pearson coefficients, all values near 0. No variable pair shows strong association.

Bar Plot



Average reviews by neighbourhood group and room type, Manhattan and Entire home/apt listings dominate review counts.

Scatter Plot

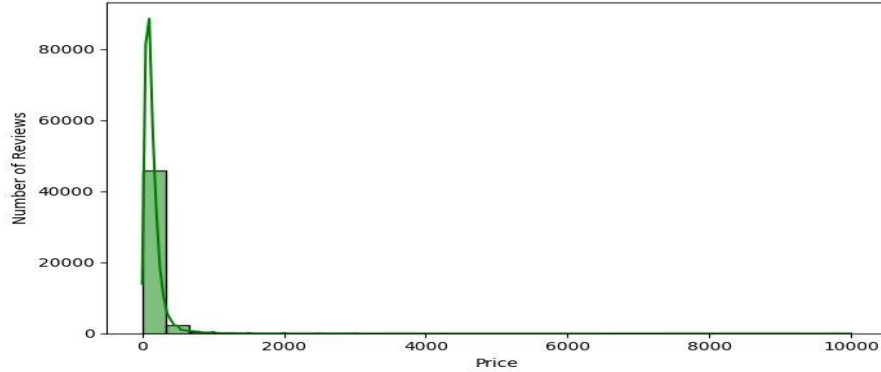


price vs availability_365 colored by cluster label, reveals the 2 natural groupings identified by KMeans.

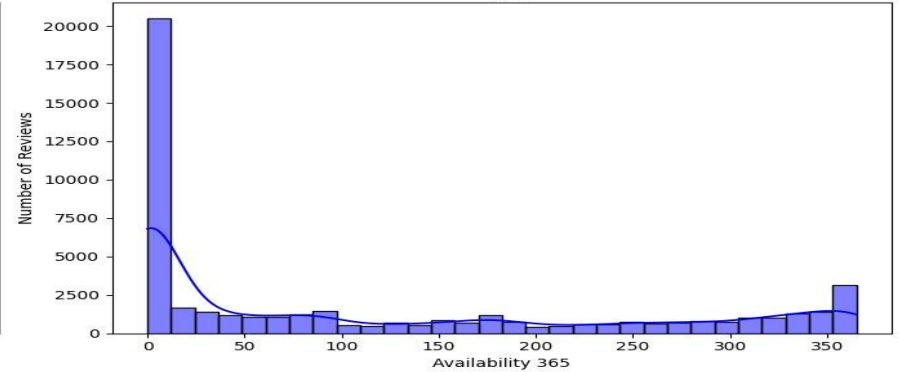
F.1 & F.2 — Histogram: Distribution of Price, Reviews & Availability

Histogram : Visualization of Distribution variables Price, Number of reviews, Availability 365

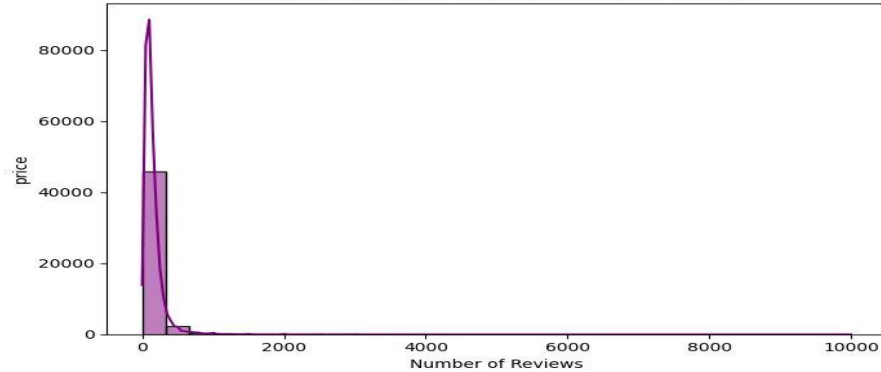
F.1 Distribution of Number of Reviews vs Price



F.2 Distribution of Availability_365 vs Number of Reviews



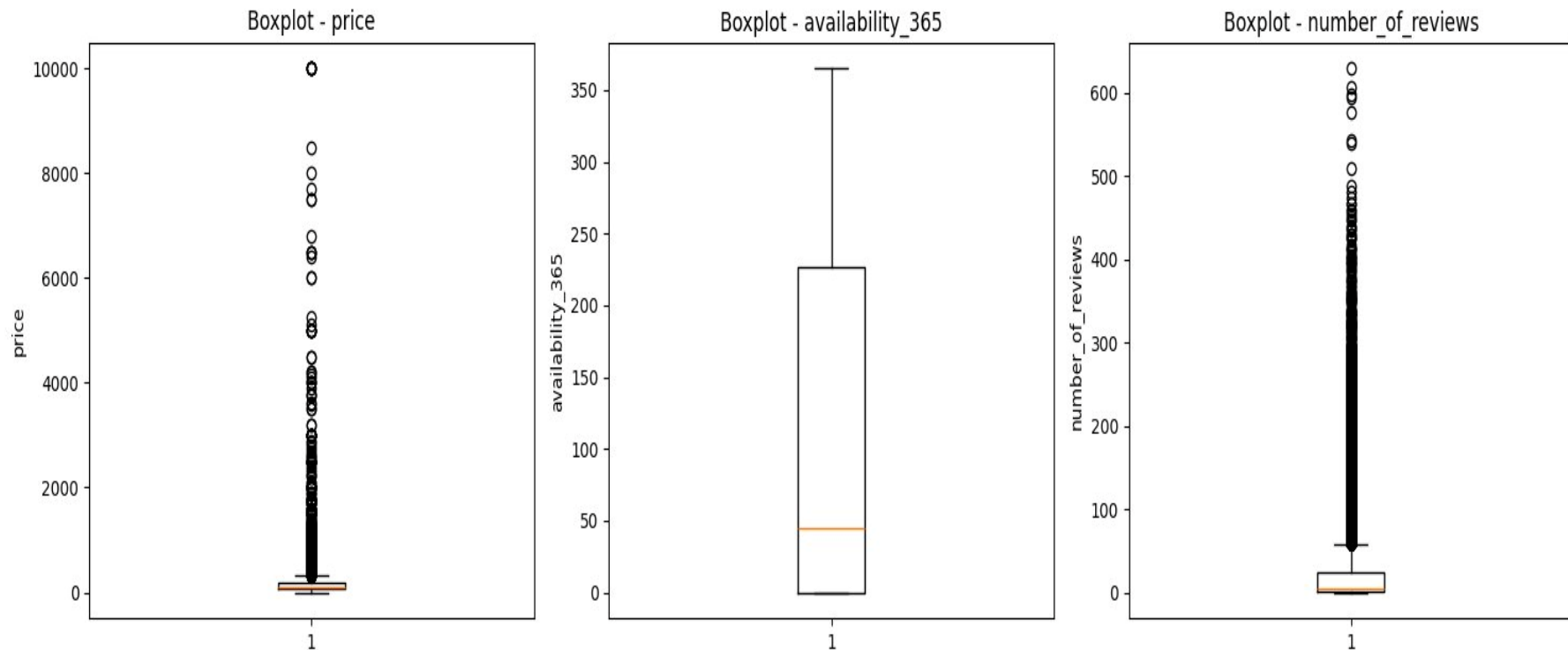
F.3.1 Distribution of Number of Reviews vs Price



All three variables are right-skewed, most listings cluster at low values with long tails confirming significant outliers.

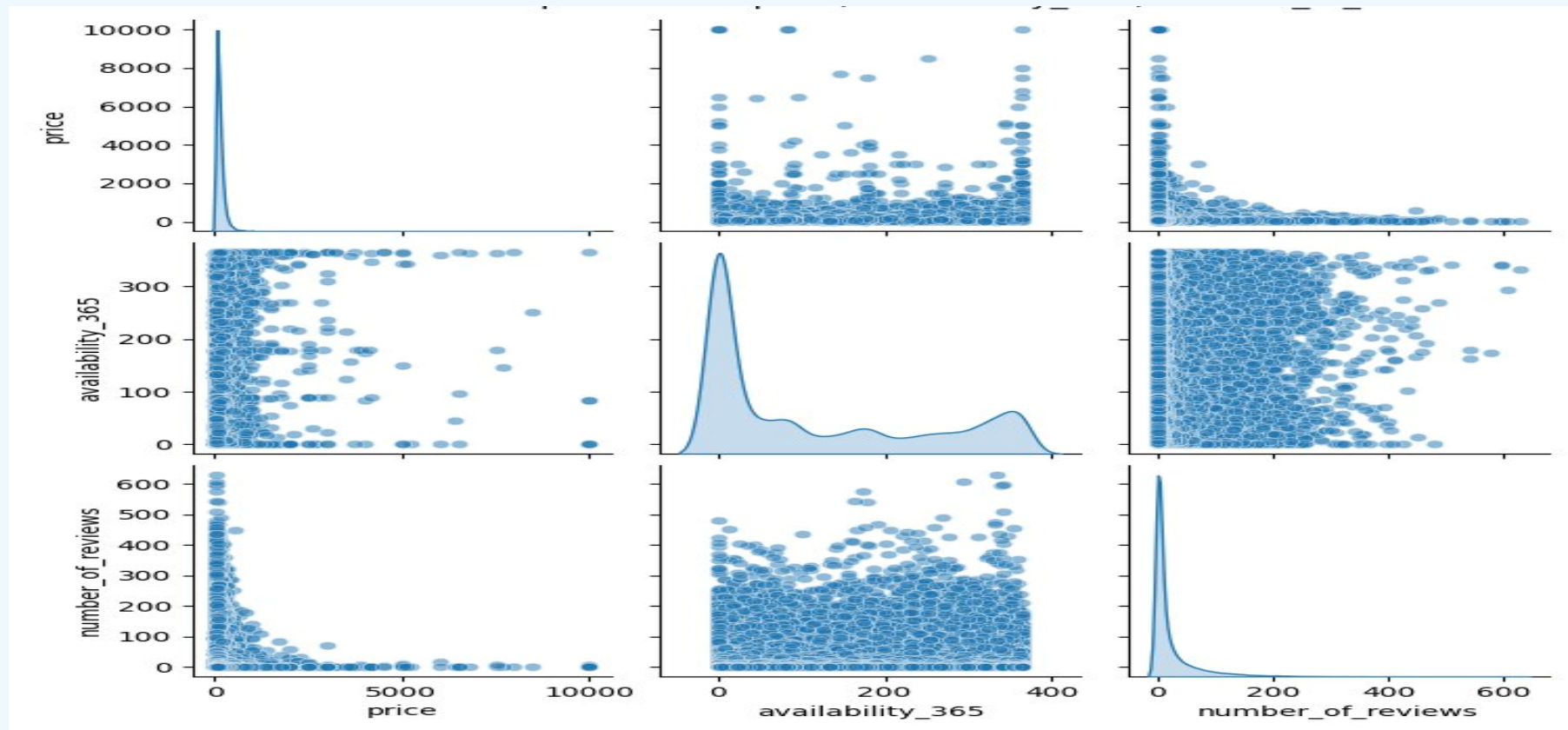
F.3.3 — Box Plot: Price, Availability 365 & Number of Reviews

F.3.3 Box Plot: Comparison between price, availability_365, number_of_reviews



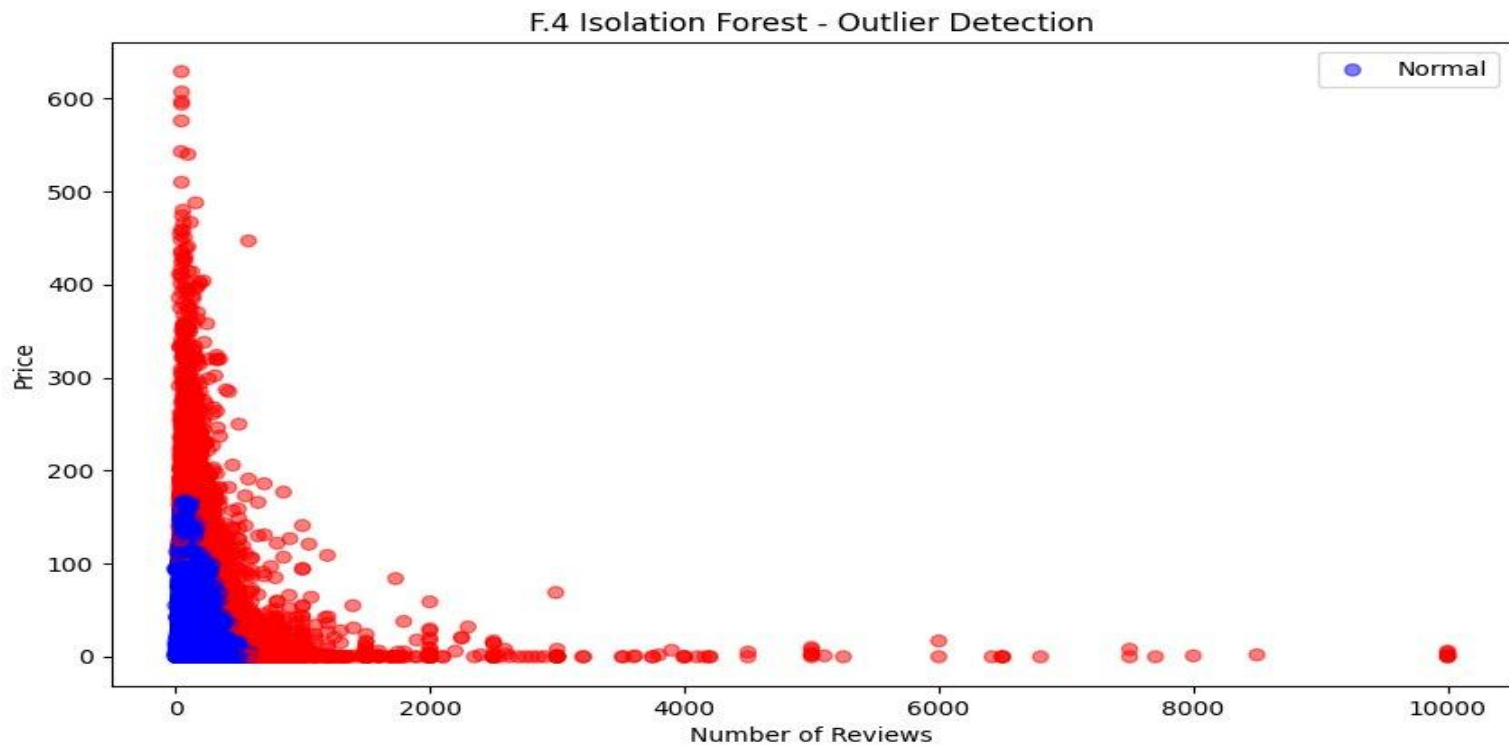
Price and reviews show many extreme dots above whiskers. Availability is more uniformly spread. Outliers are most severe in price.

Pair Plot — Relationships Between All Three Variables



No clear linear patterns visible across any pair — confirms weak correlations. Diagonal KDE shows all variables are heavily right-skewed.

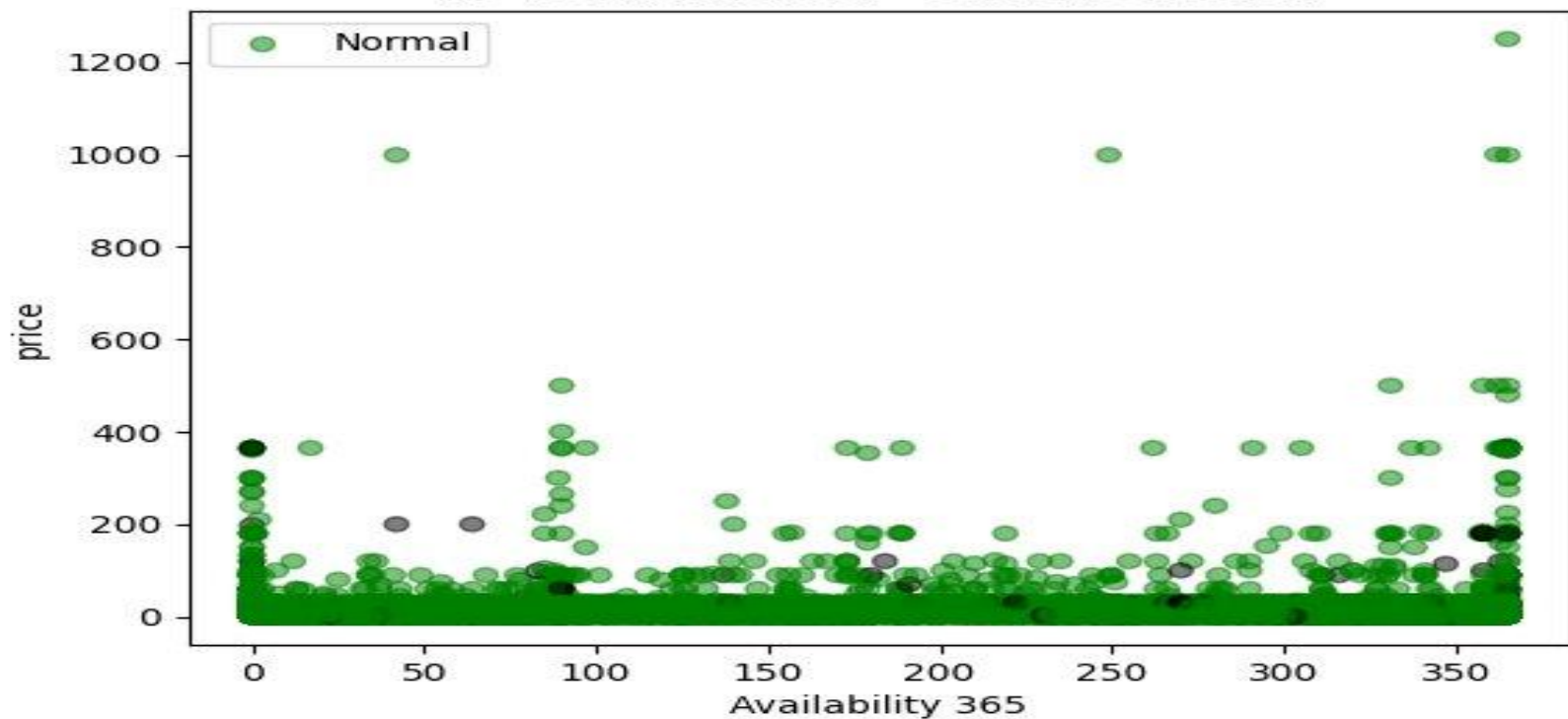
F.4 — Isolation Forest: Outlier Detection (Price vs Reviews)



Red = outliers, Blue = normal. Most outliers cluster at high review counts with low prices, and at high prices with low review counts.

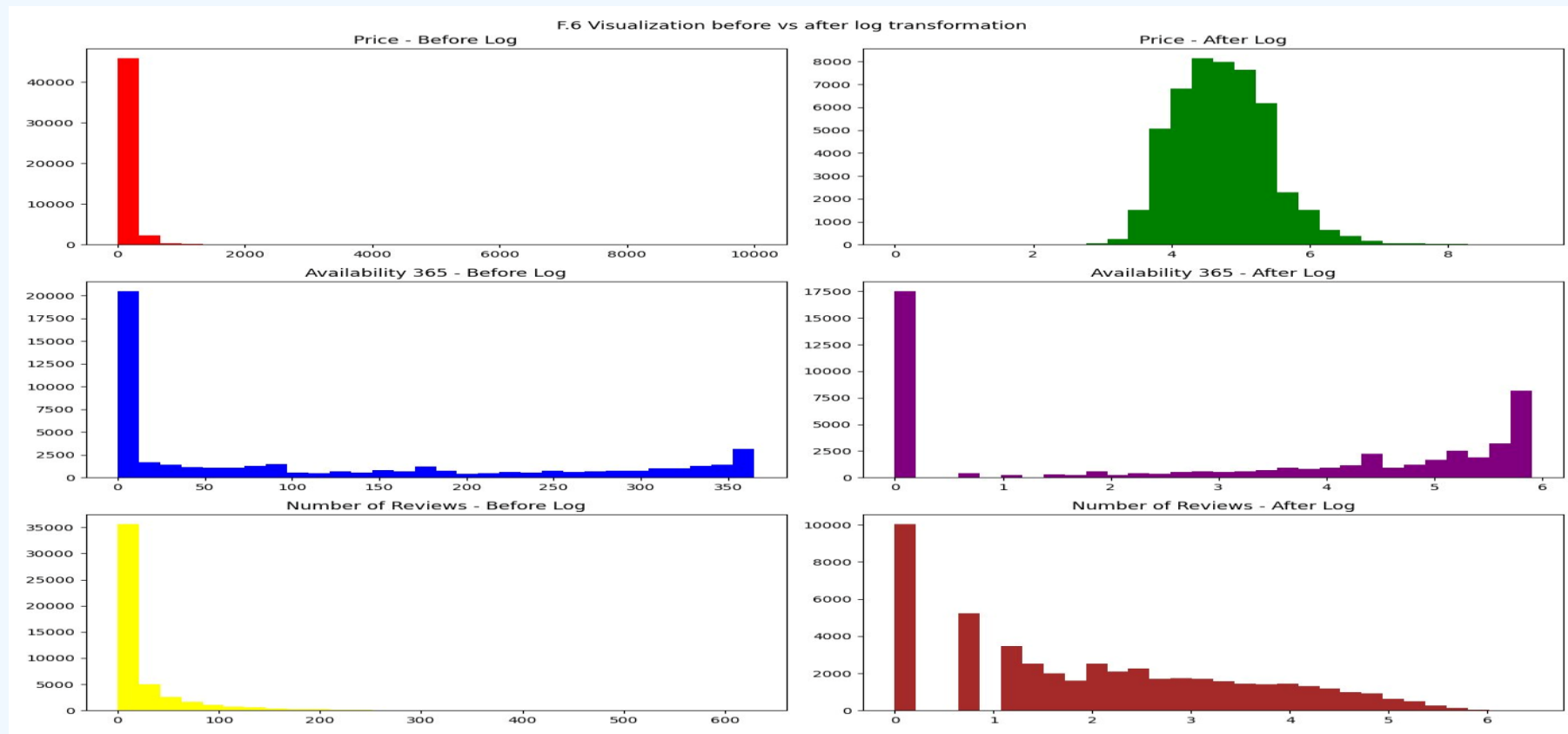
F.5 — Isolation Forest: Outlier Detection (Price vs Availability)

F.5 Isolation Forest - Outlier Detection



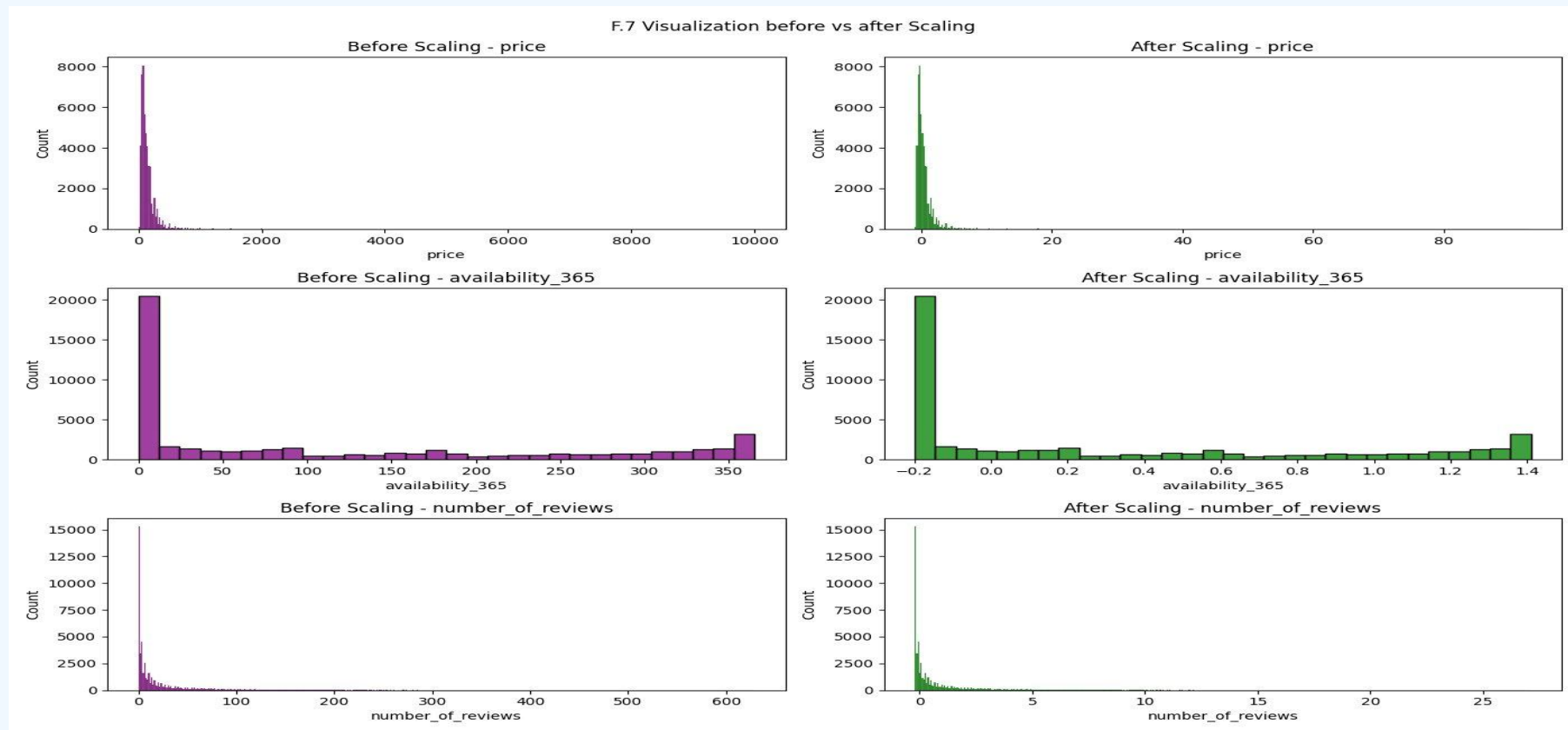
Green = normal, Dark = outliers. Extreme price listings are flagged across all availability levels. Distribution is largely normal.

F.6 — Log Transformation: Before vs After



Price skewness drops from 19.12 to 0.55. Reviews from 3.69 to 0.36. All distributions become significantly more symmetric after transformation.

F.7 — RobustScaler: Before vs After Scaling



Availability and reviews scale well. Price still shows extreme right tail even after scaling, confirming luxury listing outliers persist.

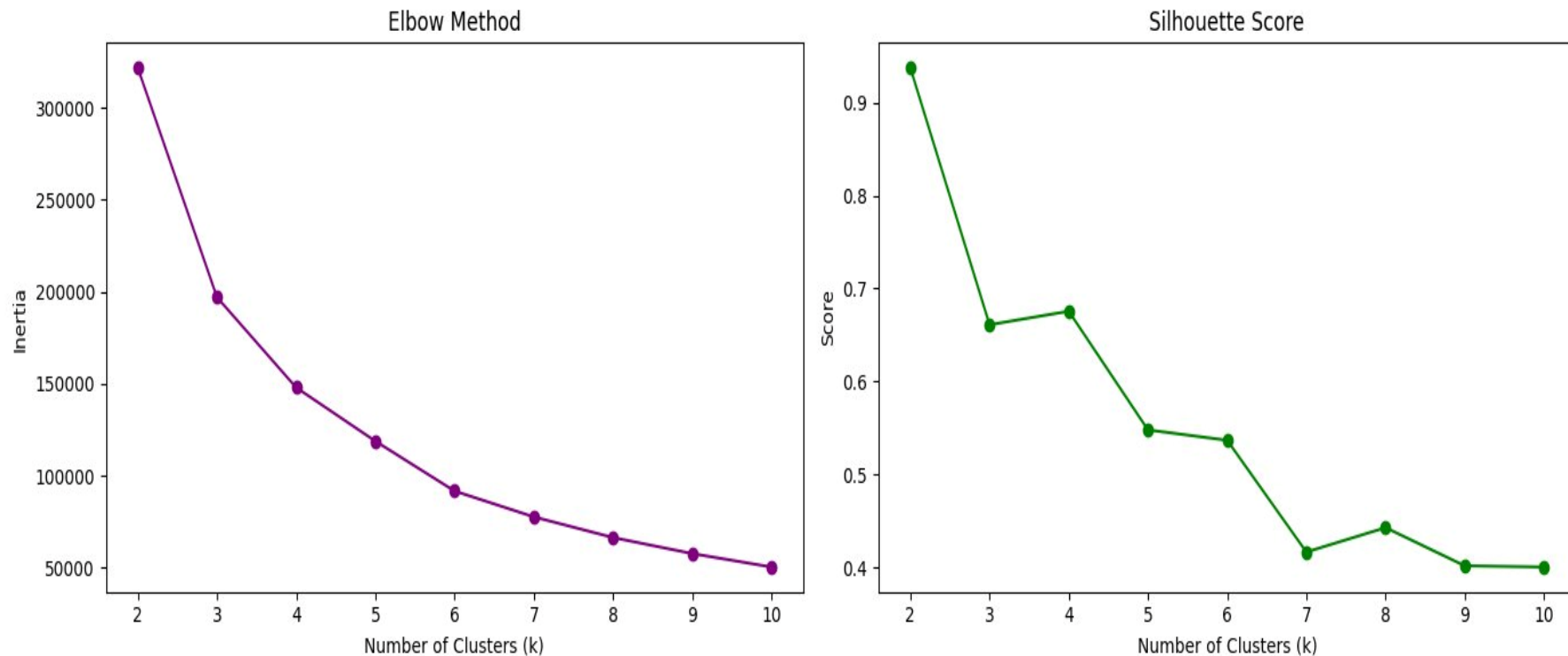
F.8 — Correlation Heatmap



All correlations near 0. Strongest pair is availability vs reviews (0.17), still very weak. Price and reviews are essentially independent (-0.048).

F.9 — Elbow Method & Silhouette Score

F.9 Application Elbow Method and Silhouette Score



Both methods confirm k=2 as optimal. Silhouette score of 0.93 at k=2 is excellent, data cleanly separates into 2 distinct clusters.

Modeling Approach

SUPERVISED LEARNING

Decision Tree Classifier

Splits data by feature thresholds to predict class labels

KNeighbors Classifier

Predicts based on k nearest training examples

Train/Test Split

80% training — 20% testing

UNSUPERVISED LEARNING

KMeans Clustering

Groups listings into k clusters by similarity

Elbow Method

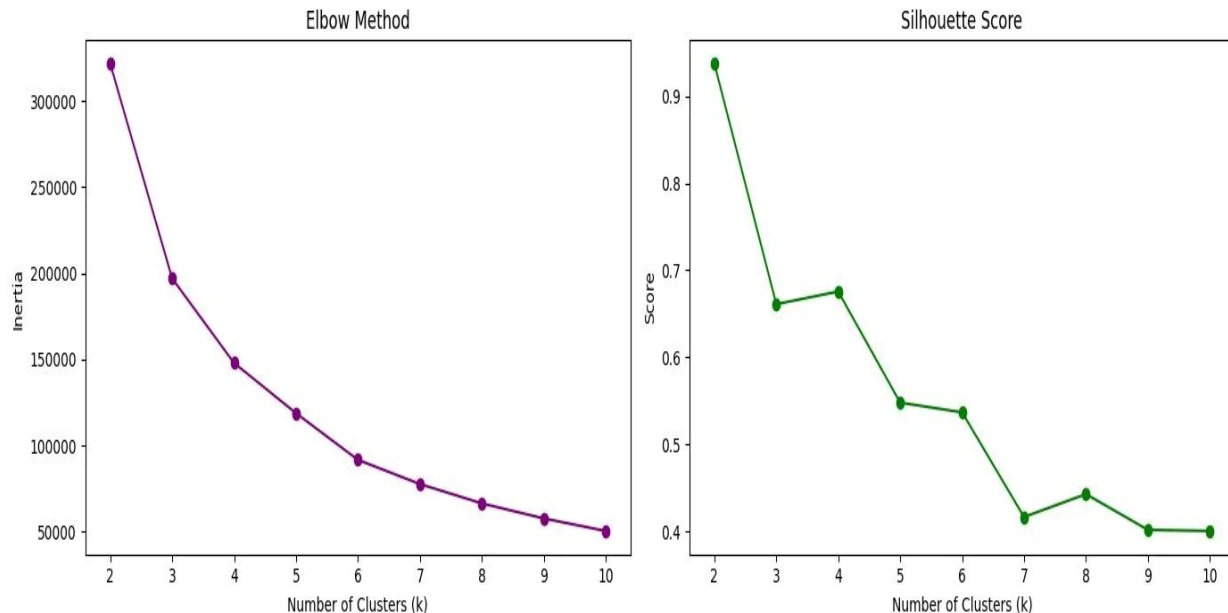
Plots inertia vs k to identify optimal clusters

Silhouette Score

Validates cluster separation quality

KMeans Clustering Results

F.9 Application Elbow Method and Silhouette Score



Best k

2

clusters

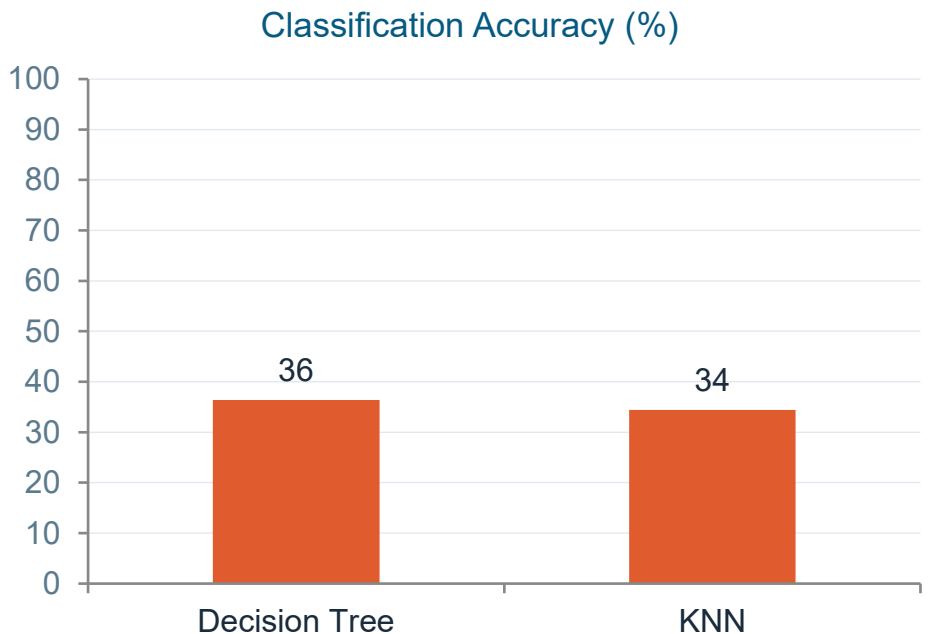
Silhouette Score

0.93

Excellent

The dataset naturally separates into 2 groups, likely budget vs premium listings

Classification Model Results



Decision Tree

36.4%

Accuracy

KNN

34.4%

Accuracy

Random guessing with 5 classes = 20%
Model accuracy only slightly above baseline

Why poor? Class imbalance + features not predictive of neighbourhood or room type

Summary of Findings



Correlation — Weak

Price, availability, and reviews are largely independent. Correlation = -0.048 (negligible). P-value significant but relationship is minimal.



Outliers — Significant

12% of data (6,021 records) flagged as outliers in `number_of_reviews`. Price had 2,972 extreme values. Availability was clean.



Skewness — Resolved

Log transformation dramatically reduced skewness in all 3 variables. RobustScaler successfully normalized the data.



Classification — Poor

Decision Tree 36.4%, KNN 34.4%. Features alone insufficient to predict neighbourhood or room type.



Clustering — Strong

KMeans identified 2 natural clusters with Silhouette Score = 0.82 (excellent). Likely budget vs premium listings.

Key Insights & Interpretation

Key Takeaway

The three variables show no meaningful linear correlation. Classification models achieved limited accuracy, suggesting these features alone are insufficient predictors. KMeans clustering revealed strong natural groupings with a silhouette score of 0.82, indicating the dataset is better suited for unsupervised analysis.

Q: Do reviews predict price?

A: No, correlation is only -0.048 . Review volume has negligible effect on price.

Q: Does availability predict reviews?

A: No, correlation is 0.017 . Hosts set availability independently of guest demand.

Q: Are these features sufficient for classification?

A: No, 36% accuracy suggests other features (amenities, location precision) are needed.

Q: What is the dataset best suited for?

A: Unsupervised analysis, the 2 natural clusters provide actionable market segmentation.

Recommendations & Next Steps

01

Expand Feature Set

Include amenities, property size, proximity to transit, and host response rate to improve classification accuracy.

02

Linear Regression

Apply $\log(\text{price})$ regression to identify which features most strongly predict listing price.

03

Time-Series Analysis

Use last_review dates to analyze seasonal pricing trends and review patterns over time.

04

Neighbourhood Deep Dive

Focus on high-growth neighbourhoods. Manhattan dominates, but outer boroughs show emerging demand.

05

Interactive Dashboard

Build a dashboard with borough/room type filters, price distributions, and cluster visualizations.

Thank You

New York City Airbnb Data Analysis

0.82

Silhouette Score
(KMeans)

-0.048

Overall
Correlation

k = 2

Optimal
Clusters

Data Source: NYC Airbnb Open Data — Kaggle (AB_NYC_2019.csv)